

Answer **all** questions.

1. (a) Carys and Pavel prepared crystals of zinc sulfate by reacting zinc carbonate with an acid.
- (i) Give the name of the acid they used. [1]
- .....
- (ii) In the first stage of their preparation, they added excess zinc carbonate to the acid.
- I. Give the observation that **immediately** shows a reaction is taking place. [1]
- .....
- II. State why they added **excess** zinc carbonate. [1]
- .....
- (iii) Describe the remaining two stages they carried out to obtain a **pure** sample of zinc sulfate crystals. [2]
- .....
- .....
- .....
- (iv) Give the chemical formula of zinc sulfate. [1]
- .....



- (b) In another experiment, Carys and Pavel investigated the temperature rise when dilute hydrochloric acid neutralises sodium hydroxide solution.



The acid was added  $5\text{ cm}^3$  at a time to  $25\text{ cm}^3$  of sodium hydroxide solution. They recorded the highest temperature reached after each addition using a digital thermometer.

They obtained the following results. The result for  $15\text{ cm}^3$  of acid is missing.

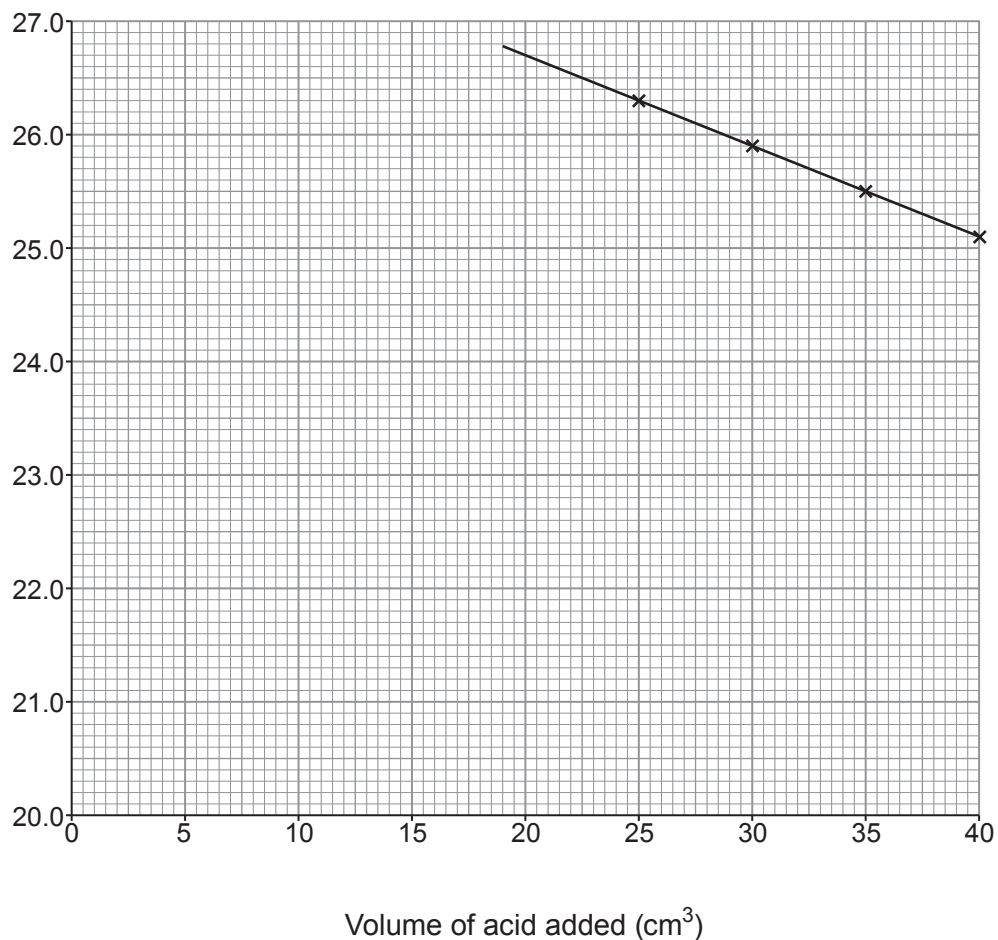
| Volume of acid added ( $\text{cm}^3$ ) | Temperature ( $^{\circ}\text{C}$ ) |
|----------------------------------------|------------------------------------|
| 0                                      | 20.4                               |
| 5                                      | 21.9                               |
| 10                                     | 23.4                               |
| 15                                     |                                    |
| 20                                     | 26.4                               |
| 25                                     | 26.3                               |
| 30                                     | 25.9                               |
| 35                                     | 25.5                               |
| 40                                     | 25.1                               |



- (i) The last four results have been plotted on the grid below and a straight line drawn through the points.

Plot the remaining four points and draw a **straight line** through them so that it intersects the line already drawn. [3]

Temperature ( $^{\circ}\text{C}$ )



- (ii) I. Use your graph to give the temperature that would have been recorded when  $15\text{ cm}^3$  of acid was added. [1]

.....  $^{\circ}\text{C}$

- II. Carys and Pavel concluded that the volume of acid needed to just neutralise all the sodium hydroxide solution was somewhere between  $20\text{ cm}^3$  and  $25\text{ cm}^3$ .

Use the graph to suggest the exact volume of acid needed. [1]

.....  $\text{cm}^3$



(iii) The temperatures recorded are slightly **lower** than expected.

Tick (✓) to show which **two** improvements to the method would enable Carys and Pavel to obtain results closer to the expected values. [2]

use a beaker instead of a flask

☐

repeat the method

☐

add the acid in smaller intervals

☐

wrap cotton wool around the flask

☐

use a larger flask

☐

place a lid on the flask

☐

(c) In a different experiment, it was found that the maximum temperature was reached when  $40\text{ cm}^3$  of hydrochloric acid was added to  $20\text{ cm}^3$  of sodium hydroxide solution.

State what this means in terms of the relative concentrations of the solutions used. [2]

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